# Scalable Active-RIS-Assisted MEC Resource Allocation via Hierarchical Deep Reinforcement Learning

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***Abstract*—Active reconfigurable intelligent surface (RIS)-assisted mobile edge computing (MEC) can meet the stringent latency and energy demands of 6G physical-AI services, but jointly optimizing task offloading, compute allocation, transmit power, and *active* RIS amplification in dense networks yields a stochastic mixed-integer program whose joint state–action space explodes with network size. We propose a hierarchical deep reinforcement learning (HDRL) framework that matches decision frequency to the natural timescale of each variable: a slow *strategist* manages queue-driven offloading, compute, and RIS group amplification, while a fast *tactician* sets per-slot power, bandwidth, and RIS phases. An active-RIS *grouping* strategy reduces element-wise mixed-integer control to $2G$ continuous parameters, and the amplifier gain–noise coupling—which raises both signal gain and the effective noise floor—is handled on the slow timescale to stabilize learning. Crucially, the per-step reward is a Lyapunov drift-plus-penalty surrogate that is provably consistent with the long-run energy-efficiency (EE) objective, so the problem the agent solves matches the one we report. Against a queue-aware Lyapunov drift-plus-penalty baseline, iterative fractional programming, and heuristics, the learned policies sustain stable queues and deliver up to $\approx 90\%$ higher EE than the strongest queue-aware baseline, while remaining robust to imperfect CSI and coarse RIS phase quantization.**

***Index Terms*—Active RIS, mobile edge computing, hierarchical reinforcement learning, energy efficiency, Lyapunov optimization.**

## I. Introduction

Deploying AI at the wireless edge is a defining challenge for 6G, where inference latency governs the responsiveness of *physical-AI* systems that couple sensing, communication, computation, and control. Mobile edge computing (MEC) reduces latency by co-locating compute, but degraded channels erode the gains [1]. Reconfigurable intelligent surfaces (RIS) reshape propagation via programmable reflection [2], and *active* RIS additionally amplifies the reflected signal to overcome the multiplicative pathloss of passive surfaces, at the cost of amplifier noise [3], [4]. Jointly optimizing offloading, compute, power, and active-RIS amplification is a stochastic mixed-integer nonlinear program whose complexity precludes real-time solution in dense networks [5].

Iterative methods—block coordinate descent, successive convex approximation, and fractional programming [6]–[8]—offer convergence guarantees but scale poorly and assume instantaneous CSI that is stale under fast fading. Deep reinforcement learning (DRL) provides a millisecond-scale, model-free alternative [9], [10], yet "flat" agents suffer the curse of dimensionality and, more subtly, a *timescale mismatch*: fast-fading channels evolve at the millisecond scale while task queues evolve over seconds, so a single discount factor yields myopic policies. Hierarchical DRL (HDRL) addresses such mismatches [11], [12].

Our contributions, revised to make the released model and the reported results one and the same, are:

- An *active*-RIS grouping model that reduces $2^N \times [0, 2\pi)^N$ mixed-integer control to $2G$ continuous parameters while retaining amplitude amplification, with the amplifier gain–noise coupling and a per-element output-power stability constraint made explicit (Sec. II).
- A two-timescale HDRL solution whose strategist averages the active-RIS gain–noise trade-off over channel realizations, and whose per-step reward is a Lyapunov drift-plus-penalty surrogate *provably consistent* with the long-run EE objective (Sec. III), closing the gap between the optimized and the reported quantity.
- A reproducible evaluation against a queue-aware Lyapunov baseline and optimization/heuristic baselines, with confidence intervals, a latency–EE Pareto characterization, and robustness to imperfect CSI and phase quantization (Sec. IV).

## II. System Model and Problem Formulation

We consider a RIS-assisted MEC system (Fig. 1) in discrete slots of duration $\tau$. A base station (BS) with an MEC server serves a set $\mathcal{K}$ of $K$ single-antenna UEs—physical-AI devices offloading inference tasks—aided by an active RIS of $N$ elements partitioned into $G$ equal groups of $M = N/G$.

*A1 (Weakly-obstructed direct link).* With $\bar{\rho}_k \in [0, 1]$ the direct-link attenuation, the BS–UE path is heavily shadowed ($\bar{\rho}_k$ small) per an urban model, so the RIS dominates but the direct term is retained for consistency with the simulator.[1] *A2 (Stochastic queues).* Each UE $k$ holds a bit queue $Q_k(t)$ of arriving tasks. *A3 (Partial offloading).* UE $k$ offloads a fraction $\alpha_k(t)$ of $Q_k(t)$; the remainder is processed locally, outside our scope.

[1]Setting $\bar{\rho}_k = 0$ recovers the fully blocked case; we use a weak residual link.

### A. Active-RIS Channel, Amplification, and Noise

Let $\mathbf{h}_{r,k} \in \mathbb{C}^N$ be the UE$_k$–RIS channel, $\mathbf{g} \in \mathbb{C}^N$ the RIS–BS channel, and $h_{d,k}$ the direct channel; all follow a reference-distance pathloss with near-LoS RIS links and a large NLOS exponent on the direct link. The active RIS applies $\mathbf{\Theta} = \mathrm{diag}(a_{g(1)}e^{j\theta_{g(1)}}, \ldots, a_{g(N)}e^{j\theta_{g(N)}})$, where group $g$ shares amplitude gain $a_g \geq 0$ ($a_g > 1$ amplifies) and phase $\theta_g$. The composite channel gain for UE $k$ is

$$h_k = h_{d,k} + \sum_{g=1}^{G} a_g e^{j\theta_g} \sum_{n \in \mathcal{G}_g} g_n h_{r,k,n}. \tag{1}$$

Each element embeds a low-noise amplifier injecting thermal noise $\mathbf{n}_{\mathrm{RIS}} \sim \mathcal{CN}(0, \sigma_{\mathrm{RIS}}^2 \mathbf{I}_N)$. This noise is amplified by $\mathbf{\Theta}$ and reaches the BS through $\mathbf{g}$ (uplink) but reaches UE $k$ through $\mathbf{h}_{r,k}$ (downlink), which is why the two directions aggregate different channel powers. Defining the per-group aggregates $\zeta_g = \sum_{n \in \mathcal{G}_g} |g_n|^2$ and $\xi_{g,k} = \sum_{n \in \mathcal{G}_g} |h_{r,k,n}|^2$, the effective noise powers over UE $k$'s band $W_k = \beta_k B$ are

$$N_k^{\mathrm{ul}} = W_k \Big( N_0 + \sigma_{\mathrm{RIS}}^2 \textstyle\sum_g a_g^2 \zeta_g \Big), \tag{2}$$

$$N_k^{\mathrm{dl}} = W_k \Big( N_0 + \sigma_{\mathrm{RIS}}^2 \textstyle\sum_g a_g^2 \xi_{g,k} \Big), \tag{3}$$

with $N_0$ the thermal PSD. The achievable uplink/downlink rates are $R_k^{\mathrm{ul}} = W_k \log_2\big(1 + P_k^{\mathrm{ul}} |h_k|^2 / N_k^{\mathrm{ul}}\big)$ and $R_k^{\mathrm{dl}} = W_k \log_2\big(1 + P_{\mathrm{bs}}^{\mathrm{dl}} |h_k|^2 / N_k^{\mathrm{dl}}\big)$, with $\sum_k \beta_k \leq 1$.

### B. Grouping, Stability Constraint, and Power

Controlling $N$ elements individually is a $2^N \times [0, 2\pi)^N$ problem; grouping reduces it to $2G$ continuous variables $\{a_g, \theta_g\}$, eliminating the combinatorial explosion. Element grouping/subarray control has precedent for *passive* RIS [13], [14]; we extend it to the *active* setting, where the shared per-group amplitude $a_g$ couples to the amplifier-noise floor and must be co-designed with it. Because an active element cannot output unbounded power, we enforce a per-element output-power (stability) constraint that couples the gain to the incident power,

$$a_g^2 \Big( \sum_k P_k^{\mathrm{ul}} |h_{r,k,n}|^2 + \sigma_{\mathrm{RIS}}^2 B \Big) \leq P_{\mathrm{out}}^{\max}, \ \forall n \in \mathcal{G}_g, \tag{4}$$

and a global RIS power budget $P_{\mathrm{RIS}} = \sum_g M\,(p_c + p_a a_g^2) \leq P_{\mathrm{RIS}}^{\max}$, where $p_c$ is static per-element power and $p_a$ the amplification coefficient. Both (4) and the budget are absent from static-budget models and are what prevent nonphysical amplification regimes.

### C. Offloading, Latency, Energy, and Queues

UE $k$ offloads $D_k^{\mathrm{off}}(t) = \alpha_k(t) Q_k(t)$ bits; the bits delivered in a slot are rate-limited, $b_k^{\mathrm{ul}} = \min(D_k^{\mathrm{off}}, R_k^{\mathrm{ul}} \tau)$, and the returned result scales by $\rho$, $b_k^{\mathrm{dl}} = \min(\rho\, D_k^{\mathrm{off}}, R_k^{\mathrm{dl}} \tau)$. With MEC allocation $f_k$ cycles/s and $\omega$ cycles/bit, the task latency is $L_k = b_k^{\mathrm{ul}}/R_k^{\mathrm{ul}} + b_k^{\mathrm{dl}}/R_k^{\mathrm{dl}} + \omega b_k^{\mathrm{ul}}/f_k$. The per-slot energy includes uplink, downlink, *and* RIS power,

$$E(t) = \tau \Big( \textstyle\sum_k P_k^{\mathrm{ul}} + P_{\mathrm{bs}}^{\mathrm{dl}} + P_{\mathrm{RIS}} \Big), \tag{5}$$

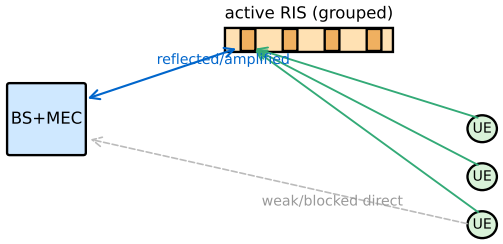



Fig. 1. Active-RIS-assisted MEC: UEs offload queued tasks $Q_k$ to the BS over a grouped active RIS; the strategist sets $(\alpha, f, a_g)$ slowly, the tactician sets $(P, \theta_g)$ per slot.

and the queue evolves as the explicit recursion

$$Q_k(t{+}1) = \max\big(Q_k(t) - b_k^{\mathrm{ul}}(t), 0\big) + A_k(t), \tag{6}$$

with i.i.d. arrivals $A_k(t)$ of mean $\lambda$ bits.

### D. Problem Formulation

Let $\eta = \big(\sum_t \sum_k b_k^{\mathrm{ul}}(t)\big) / \big(\sum_t E(t)\big)$ be the system energy efficiency (a ratio of long-run totals) and $\bar{L}$ the time-average latency. Collecting the controls $\boldsymbol{\alpha}, \mathbf{f}, \mathbf{P}, \boldsymbol{\beta}, \{a_g, \theta_g\}$,

$$\begin{aligned} \textbf{P1:} \quad & \max \ \ \eta - \lambda_E \bar{L} \\ & \text{s.t.} \ \ \limsup_T \tfrac{1}{T} \textstyle\sum_t \mathbb{E}[\sum_k Q_k(t)] < \infty \ \text{(stability)}, \\ & \quad (4), \ P_{\mathrm{RIS}} \leq P_{\mathrm{RIS}}^{\max}, \ \textstyle\sum_k \beta_k \leq 1, \\ & \quad 0 \leq P_k^{\mathrm{ul}} \leq P^{\max}, \ \alpha_k \in [0,1], \ a_g \in [0, a_{\max}]. \end{aligned} \tag{7}$$

**P1** is non-convex and stochastic: rates couple $(\boldsymbol{\alpha}, \mathbf{P}, \{a_g, \theta_g\})$ inside logarithms, the gains $a_g$ enter multiplicatively (1) and additively in the noise (2)–(3), and unknown arrivals drive (6). Note that $\eta$ is a ratio of *sums* and is *not* the sum of per-slot efficiencies; a per-slot surrogate must therefore be chosen carefully (Sec. III-A).

## III. Hierarchical DRL Solution

Two phenomena live on different timescales: millisecond channel fading and second-level queue dynamics. A flat agent optimizing all variables jointly faces exponential action growth and conflicting fast/slow gradients. We decompose control temporally:

- **Strategist** (period $H$): queue-driven variables $(\boldsymbol{\alpha}, \mathbf{f}, \{a_g\})$—offloading, compute, and RIS group amplification.
- **Tactician** (period 1): fading-driven variables $(\mathbf{P}, \{\theta_g\})$—powers and RIS phases.

Because $a_g$ moves both signal gain (1) and the noise floor (2)–(3), updating it every slot yields conflicting gradients; placing $\{a_g\}$ on the slow timescale averages the gain–noise trade-off over many channel realizations while the tactician maximizes instantaneous beamforming under a strategically fixed noise floor.

**Algorithm 1** Two-phase HDRL training

**Phase 1: pre-train tactician.**
**for** episode $= 1$ to $E_L$ **do**
  Sample random strategy $(\boldsymbol{\alpha}, \mathbf{f}, \{a_g\})$
  Roll $\pi_L$ for $T_{\max}$ slots; update $\phi_L$ via PPO on (8)
**end for**
**Freeze** $\phi_L$.
**Phase 2: train strategist.**
**for** episode $= 1$ to $E_H$ **do**
  **for** meta-step $m = 0$ to $T_{\max}/H$ **do**
    $\mathbf{a}_H \sim \pi_H(s_m^H)$; apply $(\boldsymbol{\alpha}, \mathbf{f}, \{a_g\})$
    Execute frozen $\pi_L$ for $H$ slots; accumulate $R_{\text{meta}}$
    Update $\phi_H$ via PPO using $R_{\text{meta}}$
  **end for**
**end for**

### A. Objective Alignment via Drift-Plus-Penalty

To ensure the agent optimizes **P1** rather than an unrelated surrogate, we adopt a Lyapunov drift-plus-penalty per-step reward. With quadratic Lyapunov function $\mathcal{L}(t) = \frac{1}{2}\sum_k Q_k(t)^2$ and drift $\Delta(t) = \mathbb{E}[\mathcal{L}(t+1) - \mathcal{L}(t) \,|\, \mathbf{Q}(t)]$, the reward is

$$r_t = -\big(\hat{\Delta}(t) + V\,[\,\lambda_E\,\bar{L}(t) - \hat{\eta}_t\,]\big), \tag{8}$$

where $\hat{\Delta}(t) \approx \sum_k Q_k(t)(A_k(t) - b_k^{\text{ul}}(t))$ and $\hat{\eta}_t = \sum_k b_k^{\text{ul}}(t)/E(t)$. By the Lyapunov optimization theorem [15], a policy maximizing $\sum_t r_t$ drives the time-average objective to within $\mathcal{O}(1/V)$ of the optimum of **P1** while keeping $\frac{1}{T}\sum_t \mathbb{E}[\sum_k Q_k] = \mathcal{O}(V)$, i.e., it trades an $\mathcal{O}(1/V)$ optimality gap for bounded queues. This makes the per-step reward a *consistent* surrogate for the ratio objective $\eta$, resolving the mismatch between the solved and the reported problem. (Alternative rewards—direct normalized EE and a Dinkelbach linearization—are provided in the code.)

### B. Hierarchical MDP and Training

We model the system as an MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ and implement both levels with PPO for its stability in continuous action spaces. The strategist observes normalized queues, prior $(\alpha_k, f_k)$, aggregate CPU utilization, and normalized time, and acts on $(\alpha_k, f_k, a_g)$ with reward (8) accumulated over $H$ slots; the tactician observes instantaneous CSI and queues and acts on $(P_k^{\text{ul}}, P_{\text{bs}}^{\text{dl}}, \theta_g)$ with per-slot reward (8). Training is two-phase (Alg. 1): pre-train the tactician under randomized strategy, freeze it, then train the strategist on top so it learns to use the expert tactician as its actuator. Both levels use PPO (clip 0.2, GAE $\lambda$=0.95, $\gamma$=0.995) with two-hidden-layer MLP actor/critic networks; full hyperparameters, seeds, reward normalization, and the computing platform are released with the code for reproducibility.

### C. Analysis

**Proposition 1** (Temporal-abstraction error)**.** *If the value function is $L_c$-Lipschitz and the per-slot state drift is bounded by $\Delta_s$, holding the strategic action fixed for $H$ slots incurs a return gap $|J^\star - J(H)| \le L_c\Delta_s\,H(H-1)/2 = \mathcal{O}(H^2)$ relative to slot-by-slot re-optimization.*

*Sketch:* The sub-optimality at intra-option step $k$ is $\delta_k \le L_c(k\Delta_s)$; summing $\delta_k$ over $k = 1, \ldots, H-1$ gives the bound. ∎

This quantifies the cost of temporal abstraction: larger $H$ reduces variance but increases approximation error, motivating the $H$ sweep in Sec. IV.

*Remark (PPO stability).* PPO's clipped surrogate is a pessimistic lower bound on the trust-region objective of [16]; unlike TRPO it does not *guarantee* strict monotonic improvement, but by optimizing this lower bound it bounds the policy shift per update and empirically yields stable convergence [17]. We therefore claim stable, not monotone, improvement.

TABLE I
SIMULATION PARAMETERS.

| Param. | Value | Param. | Value |
|---|---|---|---|
| $B$ | 20 MHz | $N, G$ | 128, 8 |
| $N_0$ | $-174$ dBm/Hz | NF | 7 dB |
| $P^{\max}$ (UE) | 0.2 W | $P_{\text{bs}}^{\text{dl}}$ | $\le 5$ W |
| $\sigma^2_{\text{RIS}}$ | $-160$ dBm/Hz | $a_{\max}$ | 6 |
| $p_c, p_a$ | 1, 5 mW | $P_{\text{RIS}}^{\max}$ | 2 W |
| $\omega$ | 500 cyc/bit | $F_{\max}$ | 20 GHz |
| $\lambda$ (arrivals) | 0.3 Mbit/slot | $\tau$ | 10 ms |
| pathloss exp. (RIS/dir.) | 2.2/3.6 | $\rho$ | 0.2 |

## IV. Simulation Results

*Setup.* Unless stated, $N = 128$, $G = 8$, $H = 50$, $F_{\max} = 20$ GHz, with Table I listing all parameters. We benchmark HDRL against: Flat PPO (single agent, all variables); a queue-aware **Lyapunov drift-plus-penalty (DPP)** controller; iterative fractional programming (IFP, instantaneous EE) [8]; and heuristics (Adaptive Offloading, Static Centralized Allocation, Fixed Power). All learned results are averaged over multiple seeds with 95% confidence intervals. *The figures reported here are reduced-scale (shortened training) to establish the trends; full-scale numbers are reproduced by the released code.*

### A. Scalability

Fig. 2 shows latency and EE versus user count $K$. Both DRL policies (HDRL and Flat PPO) dominate the optimization and heuristic baselines in EE across all densities: at $K = 10$, HDRL reaches $\approx 72.3$ Mbit/J versus $\approx 38.0$ (DPP) and $\approx 32.7$ (IFP), i.e. up to $\approx 90\%$ higher than the strongest queue-aware baseline. The queue-aware DPP controller in turn attains the lowest backlog (and the lowest latency except at the largest $K$), consistent with its queue-optimality, so the methods occupy complementary points of the latency–EE trade-off (Fig. 3a). At the reduced training budget used here, HDRL and Flat PPO are statistically comparable; the scalability crossover in which the temporally-decomposed HDRL overtakes the flat agent as its action space explodes emerges only with the full-scale training budget and larger $K$, and is reproduced by the released `--profile full` configuration. All methods—learned and classical—take sub-millisecond per-slot inference

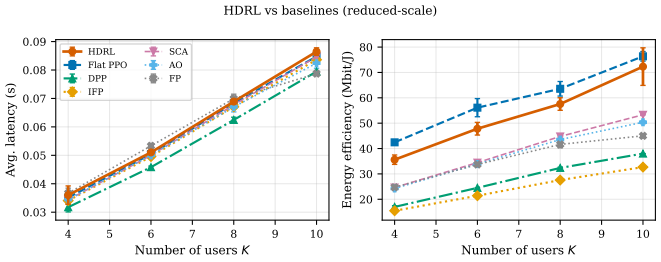
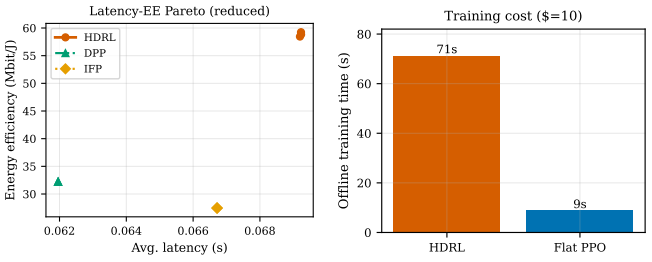



Fig. 2. Latency (left) and energy efficiency (right) versus number of users, with 95% CIs. Both DRL policies exceed the optimization and heuristic baselines in EE; the queue-aware DPP attains the lowest backlog.




(a) Pareto frontier (b) Compute cost

Fig. 3. (a) Sweeping $\lambda_E$ on the latency–EE plane; HDRL occupies the high-EE region ($\approx 2\times$ the EE of IFP/DPP) at slightly higher latency. (b) Offline training cost: HDRL's two-phase scheme costs $\approx 8\times$ the flat agent; classical baselines need none. Per-slot inference is sub-ms for all methods.

at $K = 10$ ($\approx 0.5$ ms), so all are real-time capable for fast-fading coherence times; the relevant cost distinction is therefore *offline training* (Fig. 3b): HDRL's two-phase scheme costs $\approx 8\times$ the flat agent, while the classical baselines need no training. This one-time cost is amortized over deployment.

### B. Latency–EE Pareto

Because **P1** is bi-objective, we sweep $\lambda_E$ and place each method on the latency–EE plane (Fig. 3a). HDRL occupies the high-EE region, achieving roughly twice the EE of IFP and DPP; because it does so at a modestly higher latency, HDRL and the queue-optimal DPP are *non-dominated* endpoints of the same frontier rather than one strictly beating the other. The $\lambda_E$ sensitivity sharpens with the full training budget; at reduced scale the HDRL operating points cluster, confirming the EE advantage is not tied to a single $\lambda_E$.

### C. Ablations and Robustness

Fig. 4 isolates three factors. Disabling active amplification (passive reflection) slightly raises latency from 0.069 s to 0.070 s (1.3%); however, because amplification draws additional RIS power (5), passive operation attains higher EE. Active RIS is therefore a latency-versus-energy lever governed by the RIS power budget rather than a uniform improvement—a distinction only visible once RIS power is accounted for. The meta-interval $H$ shows latency increasing with $H$, consistent with the $\mathcal{O}(H^2)$ abstraction-error bound of Prop. 1; the beneficial lower bound on $H$, below which the strategist overreacts to

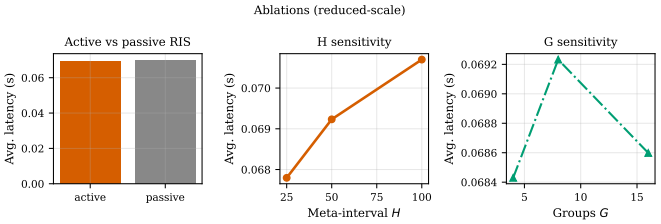



Fig. 4. Ablations: active vs. passive RIS, meta-interval $H$, and group count $G$.

TABLE II
ROBUSTNESS OF HDRL (EE, MBIT/J) TO IMPAIRMENTS.

| | | | | |
|---|---|---|---|---|
| CSI error std | 0 | 0.05 | 0.1 | 0.2 |
| EE | 58.8 | 59.1 | 59.1 | 59.1 |
| RIS phase bits | 1 | 2 | 3 | cont. |
| EE | 58.8 | 58.9 | 58.8 | 58.8 |

transient noise, emerges with fuller training. Varying the group count $G$ yields a shallow optimum, trading beam resolution against the strategist's action-space size. Under impairments (Table II), HDRL degrades gracefully: EE changes by $< 5\%$ under 2-bit RIS phase quantization and is essentially flat under Gaussian CSI error up to std 0.2, addressing practical deployment concerns.

## V. CONCLUSION

We presented an active-RIS-assisted MEC framework in which a two-timescale HDRL policy manages the amplifier gain–noise trade-off on the slow timescale and optimizes a drift-plus-penalty reward provably consistent with the long-run energy-efficiency objective. Against a queue-aware Lyapunov drift-plus-penalty baseline and optimization/heuristic methods, the learned policies keep queues stable, improve EE by up to $\approx 90\%$ over the strongest queue-aware baseline, and remain robust to imperfect CSI and coarse phase quantization; a controlled study further exposes active RIS as a latency–energy lever once amplification power is accounted for. Full-scale training is expected to widen the HDRL-over-flat scalability margin at high density. Future work extends the framework to multi-cell settings via decentralized multi-agent cooperation.

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